

AZURE AI FOUNDRY LAB - 3.2

Title :

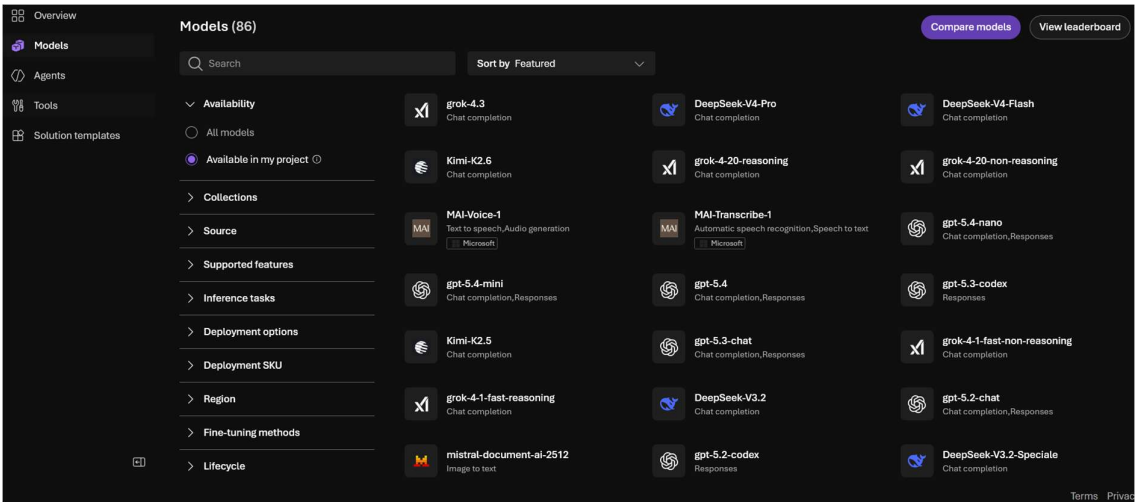
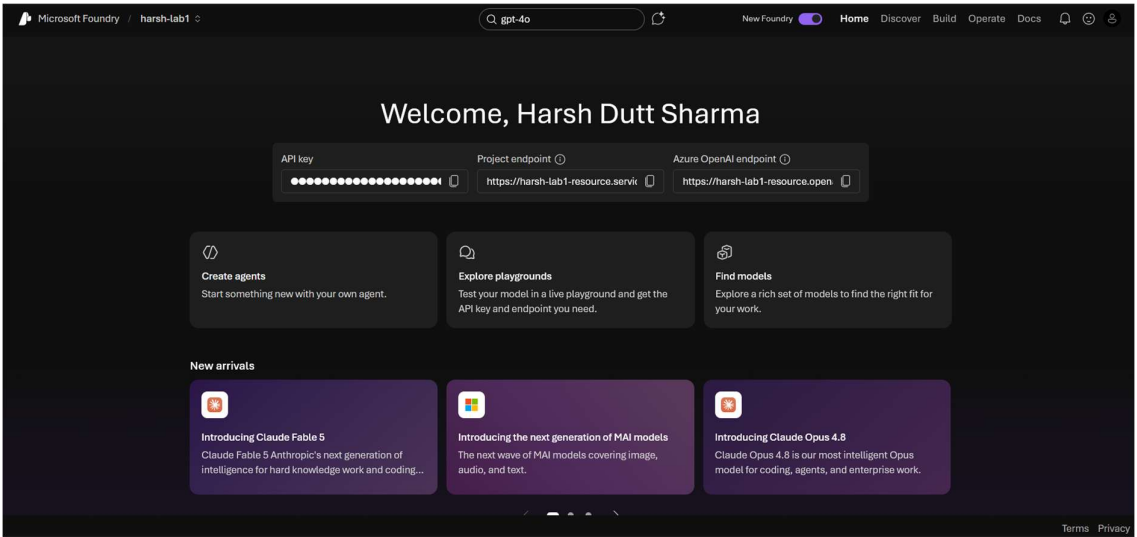
Evaluate Models from the Model Catalog

Objective :

To explore the Azure AI Foundry Model Catalog, deploy a foundation model, test its capabilities in the Playground, and evaluate model performance using Azure AI Foundry's built-in evaluation framework.

Tasks Performed :

- 1. Opened Azure AI Foundry, explored the model catalog and deployed the GPT-4.1 model and also opened the model playground.



This screenshot shows the 'gpt-4.1' model details page in the Microsoft Foundry interface. The page is divided into several sections:

- Overview:** Includes tabs for Overview, Models, Agents, Tools, and Solution templates.
- Details:** The main content area, showing the model name 'gpt-4.1' and a description: 'gpt-4.1 outperforms gpt-4o across the board, with major gains in coding, instruction following, and long-context understanding'. It also features buttons for 'Azure OpenAI' and 'Direct from Azure', and a 'Version: 2025-04-14' label.
- Quick facts:** A table summarizing key information:

Quick facts	
Model provider	Azure OpenAI
Type	Chat completion, Responses
Lifecycle	Generally Available
Input type	text, image
Output type	text
Context window	1048.576k
Token limits	32768 output
Pricing	View pricing
- Model ID:** A text box containing the identifier 'azureml://registries/azure-openai/models/gpt-4.1/versions/2025-04-14'.
- Direct from Azure models:** A section explaining that these models are a select portfolio curated for their market-differentiated capabilities, listing benefits like security, streamlined operations, and future-ready flexibility.
- Key capabilities:** A section titled 'About this model' stating that the gpt-4.1 series is the latest iteration of the gpt-4o model family, specifically targeted for better coding and instruction following.

This screenshot shows the 'gpt-4.1' playground interface. The left sidebar contains controls for the model and chat session:

- Model:** Set to 'gpt-4.1' with a dropdown for 'Global Standard deployment'.
- Instructions:** A text area containing the prompt: 'You are an AI assistant that helps people find information.'
- Tools:** A section with 'Add' and 'Upload files' buttons.
- Knowledge:** A section with a dropdown arrow.
- Memory:** A section with a 'Preview' button.

The main chat area on the right has a header with 'Chat' and 'Call model' buttons. Below the header is a large text input field with the placeholder 'What do you want to chat about?'. At the bottom, there is a text input field with the placeholder 'Chat with the model...' and a 'Send' button.

2. The deployed model was tested in the playground environment using a sample prompt.

This screenshot shows the 'gpt-4.1' playground interface with a chat session in progress. The left sidebar is identical to the previous screenshot. The main chat area shows the following:

- Chat:** A text input field with the prompt: 'Explain how renewable energy helps the environment.'
- Response:** A list of six points explaining how renewable energy helps the environment:
 - Reduces Air Pollution:** Renewable energy sources like solar, wind, and hydroelectric power produce little to no air pollution compared to burning fossil fuels (coal, oil, natural gas), which release harmful pollutants into the air.
 - Lowers Greenhouse Gas Emissions:** Using renewable energy reduces emissions of greenhouse gases such as carbon dioxide (CO₂). These gases trap heat in the atmosphere and cause global warming and climate change.
 - Decreases Water Usage:** Many renewable energy sources use significantly less water than conventional power plants. For example, solar and wind energy use very little water, conserving this vital resource.
 - Reduces Dependence on Finite Resources:** Renewable energy comes from natural sources that are constantly replenished, like sunlight and wind, so it doesn't deplete the Earth's resources.
 - Lessens Land Degradation and Habitat Loss:** Fossil fuel extraction (mining, drilling) can damage land and disrupt wildlife habitats. Renewable energy installations are often less invasive, and new technologies aim to minimize impacts further.
 - Mitigates Acid Rain and Smog:** By reducing the burning of fossil fuels, renewable energy also decreases emissions that cause acid rain and
- Input field:** A text input field with the placeholder 'Chat with the model...' and a 'Send' button.

3. A new evaluation task was created to assess the model quality and safety using Azure AI Foundry’s built-in evaluation framework.

← Create new evaluation

1 Target: Model

Tell us what you'd like to evaluate.

☐ Agent

Evaluate the quality and safety of an agent

☒ Model

Evaluate the quality and safety of a model deployment

☐ Dataset

Evaluate an existing dataset

Next

2 Data

3 Configure models

4 Criteria

5 Review

Q Search models

☒	Name	Model	Version	Capabilities	Status	Type	Date Created
☐	Phi-4	Phi-4	2	Chat Compl...	Succeeded	Base	6/13/26, 3:1...
☐	Phi-4-mini-instruct	Phi-4-mini-in...	1	Chat Compl...	Succeeded	Base	6/13/26, 5:1...
☐	gpt-4o	gpt-4o	2024-11-20	Chat Compl...	Succeeded	Base	6/19/26, 9:2...
☐	gpt-4o-mini	gpt-4o-mini	2024-07-18	Chat Compl...	Succeeded	Base	6/19/26, 9:2...
☐	gpt-4o-2	gpt-4o	2024-11-20	Chat Compl...	Succeeded	Base	6/19/26, 4:5...
☒	gpt-4.1	gpt-4.1	2025-04-14	Chat Compl...	Succeeded	Base	6/19/26, 7:5...
☐	gpt-4.1-mini	gpt-4.1-mini	2025-04-14	Chat Compl...	Succeeded	Base	6/19/26, 7:5...

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PrevNext

← Create new evaluation

Target: Model

2 Data

Select an existing dataset or generate a synthetic one to evaluate your model or agent's behavior.

Dataset source

☐ Synthetic generation

Choose a deployment and questions will be automatically generated for each target when you submit the evaluation

☒ Existing dataset

Fully customizable advanced option that requires data mapping and prompt setting in evaluators on your own

☐ Benchmarks

Industry-standard benchmarks with built-in evaluators

☐ Import chat completions

Evaluate your stored chat completions

BackNext

Select your file

Download data sampleUpload new dataset

Q Search by file name

Name	Version	Created by	Created on
☒ travel_dataset	1	Harsh Dutt Sharma	6/22/26, 6:22:47 PM
☐ gpt_4o_f70h8yt9cs	1.0	Harsh Dutt Sharma	6/19/26, 5:13:34 PM

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PrevNext

Dataset preview (Top 5 rows) ⓘ

input

What are the top tourist attractions in Paris?

What is the best time to visit Japan?

← Create new evaluation

3 Field mapping

Map your file columns to the standard evaluator fields.

Judge model *

Model: gpt-4oGlobal Standard deployment

We've auto-detected likely matches.

Query ⓘ

{{item.input}}

Response ⓘ

{{sample.output_text}}

Context ⓘ

Not available

Ground truth ⓘ

Not available

Tool calls ⓘ

Not available

Tool definitions ⓘ

Not available

BackNext

Instructions

Available fields in your file:

Input

Why field mapping now?

Evaluators need to know which part of your data is the query, so we can use them to test your targets. Mapping more fields like context and ground truth will unlock more types of evaluators in the next step.

What evaluators are not auto-selected here?

Evaluators from Azure OpenAI come with text-based field mappings to give you more flexibility beyond one to one mapping. You will need to specify field relationship when you construct your system, user and assistant prompts. Don't worry, we have templates so you won't have to start from scratch.

Add custom prompt

gpt-4.1

gpt-4.1

Temperature ⓘ

1

Top P ⓘ

1

Max Completion Tokens ⓘ

13107

Prompt *

Use {{curly braces}} to insert variables.

DEVELOPER

You are a professional travel assistant.

Answer the user's travel question directly and concisely.

Provide specific recommendations, destinations, budgets, do

Do not introduce yourself.
Do not add generic greetings.

Save

Cancel

Field mapping

Configure models

Criteria

Choose what criteria you want to evaluate — quality, risk and safety, and more.

We've auto-suggested the following 8 evaluators

Agents (3)

Remove all

TaskCompletion X IntentResolution X CustomerSatisfaction X

Quality (4)

Remove all

Groundedness X Fluency X Coherence X Relevance X

Business (1)

Remove all

DeflectionRate X

Add new evaluator

Back

Next

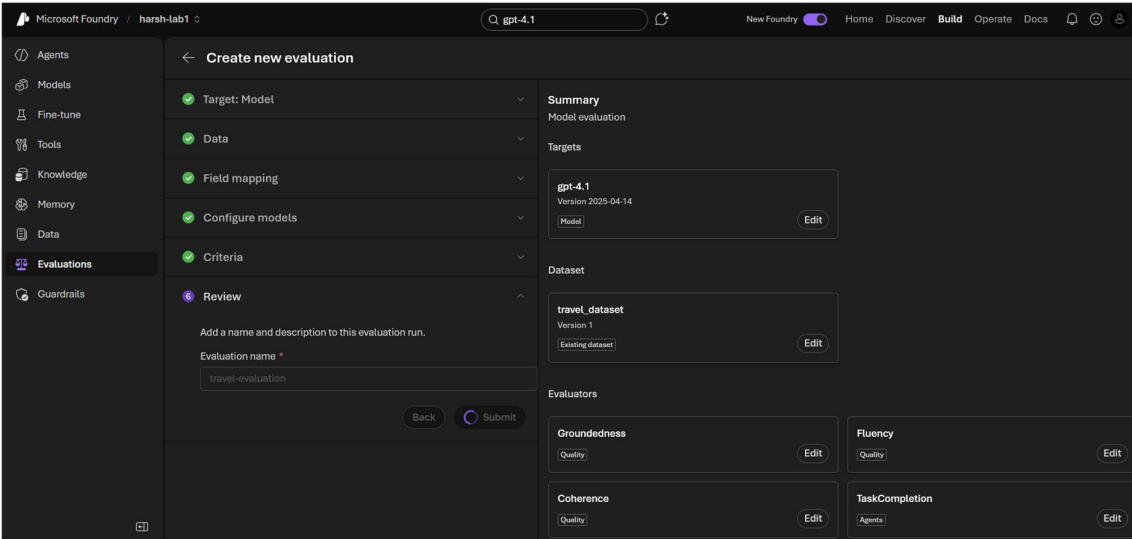
Mapped fields from your dataset:

query: {{item.input}} response: {{sample.output_text}}

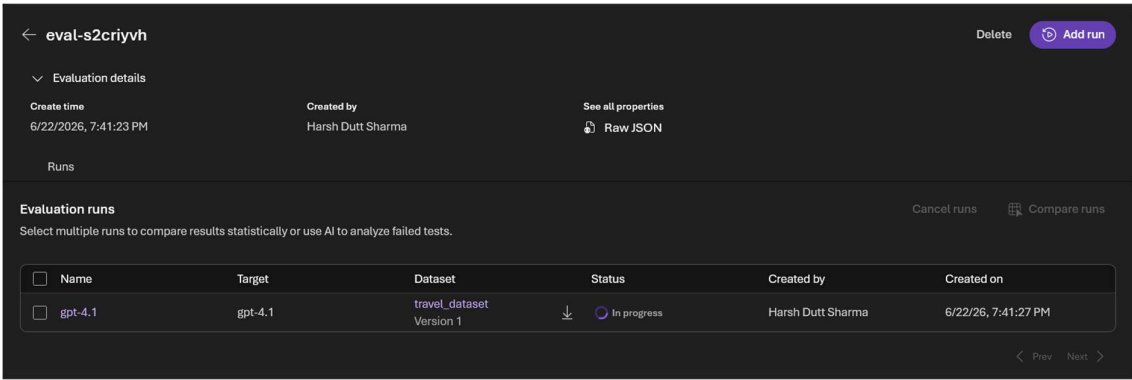
Instructions

What evaluators are not auto-selected here?

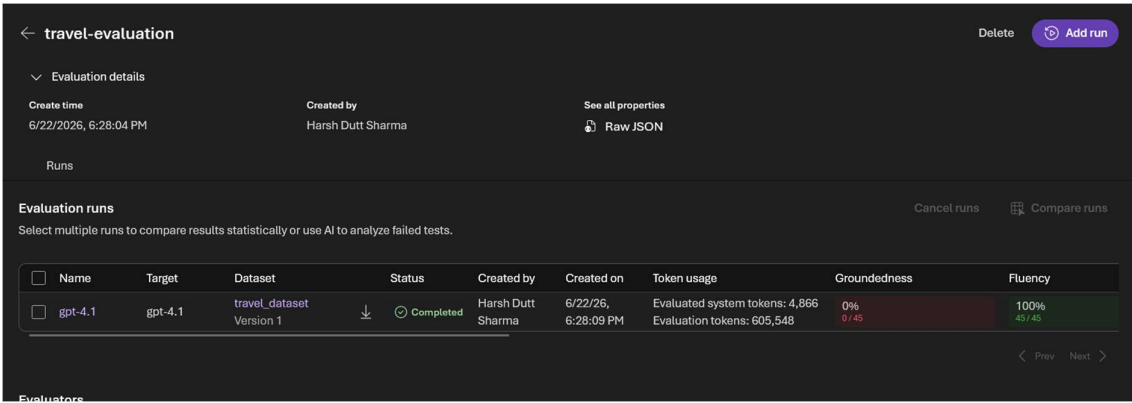
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4. Configured evaluation metrics and executed the evaluation run :



5. Successfully evaluated the GPT-4.1 model.



Coherence	TaskCompletion	Relevance	IntentResolution	CustomerSatisfaction	DeflectionRate
2% 1 / 45	0% 0 / 45	4% 2 / 45	2% 1 / 45	4% 2 / 45	100% 45 / 45

Results :

The evaluation was completed successfully and the following metrics were generated by Azure AI Foundry.

Metric	Score
Fluency	100%
Groundedness	7%
Coherence	2%
Task Completion	0%
Relevance	4%
Intent Resolution	2%
Customer Satisfaction	4%
Deflection Rate	100%

Observations :

- 1. The deployed GPT-4.1 model generated fluent and grammatically correct responses, but evaluation scores showed that response quality depends heavily on prompt design and dataset configuration.
- 2. Different evaluation metrics such as Fluency, Relevance, Coherence, and Task Completion provide insights into various aspects of model performance.
- 3. Even small changes in the system prompt can significantly affect model behavior and evaluation results, highlighting the importance of prompt engineering in Generative AI applications.

Conclusion :

This lab demonstrated the complete workflow of exploring, deploying, testing, and evaluating foundation models in Azure AI Foundry. Through hands-on experimentation with GPT-4.1 and the evaluation framework, an understanding of model behavior, prompt engineering, and performance assessment was developed. The exercise highlighted the importance of systematic evaluation in building reliable Generative AI applications.